Sampling and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Sampling and Representation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Overfitting Intuition

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Sampling and Representation trace and record the first step you can explain without help.

Use W2: Read the **sample** result for Questions **population**s and **sample**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Sampling and Representation explanation against the supplied facts before discussing it.

Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Plan independently

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Sampling and Representation target to a trace and a justified conclusion.

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Sampling and Representation case and show the deciding evidence.

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 26

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 19 Test and reflect

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe **response** before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Sampling and Representation adds to Transparent Book Recommender and keep the evidence.

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Define the population

> Define the population. Apply the supplied Sampling and Representation facts and keep your reasoning.

The question is whether all Grade 7 students prefer a longer lunch. Name the **population** and one unsuitable **sample**.

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Read the sample result

> Read the sample result. Apply the supplied Sampling and Representation facts and keep your reasoning.

Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

My reasoning and evidence

WORKBOOK · 14 OF 26

W3 Repair a claim

> Repair a claim. Apply the supplied Sampling and Representation facts and keep your reasoning.

A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

My reasoning and evidence

WORKBOOK · 15 OF 26

W4 Plan independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Plan independently. Apply the supplied Sampling and Representation facts and keep your reasoning.

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

My reasoning and evidence

W5 Separate invitation and response

Separate invitation and response. Apply the supplied Sampling and Representation facts and keep your reasoning.

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report non**response**?

My reasoning and evidence

Graphs and honest scales — W1 Read the scale

> Use the supplied Sampling and Representation task to record your reasoning, evidence and next step.

A bar is 6 squares long with 2 votes per square. How many votes is that?

My reasoning and evidence

Graphs and honest scales — W2 Compare fairly

> Use the supplied Sampling and Representation task to record your reasoning, evidence and next step.

Counts are 12 and 15. State the difference and ratio using the true values.

My reasoning and evidence

WORKBOOK · 19 OF 26

Graphs and honest scales — W3 Repair a misleading graph

> Use the supplied Sampling and Representation task to record your reasoning, evidence and next step.

Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

My reasoning and evidence

WORKBOOK · 20 OF 26

Graphs and honest scales — W4 Draw independently

> Use the supplied Sampling and Representation task to record your reasoning, evidence and next step.

Draw a **bar chart** for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero **baseline** and equal **scale**. State two supported comparisons.

My reasoning and evidence

WORKBOOK · 21 OF 26

Graphs and honest scales — W5 Limit the claim

> Use the supplied Sampling and Representation task to record your reasoning, evidence and next step.

Art has the most votes among 18 **response**s. Does that establish Art is best for learning?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Sampling and Representation

> A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Explain the labelled step before checking the changed condition.

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

AI ACADEMY / AI-2/19

Sampling and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

The question concerns all Grade 7, not only library-club members.



2. Select invitations

Teacher uses the complete approved list and documented random selection.



3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



4. Report scope

Invitation response rate 12/15=80%; preference counts describe respondents.

CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can **80% response** be reported as **80%** of the whole grade responding?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

> 1. Define target

2. Report scope

3. Record **response**s

4. Select invitations

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report scope Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Invite only the 15 library-club members instead. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The invitation **process** no longer covers the whole grade through this frame.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The invitation **process** no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Repair a missing building block

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

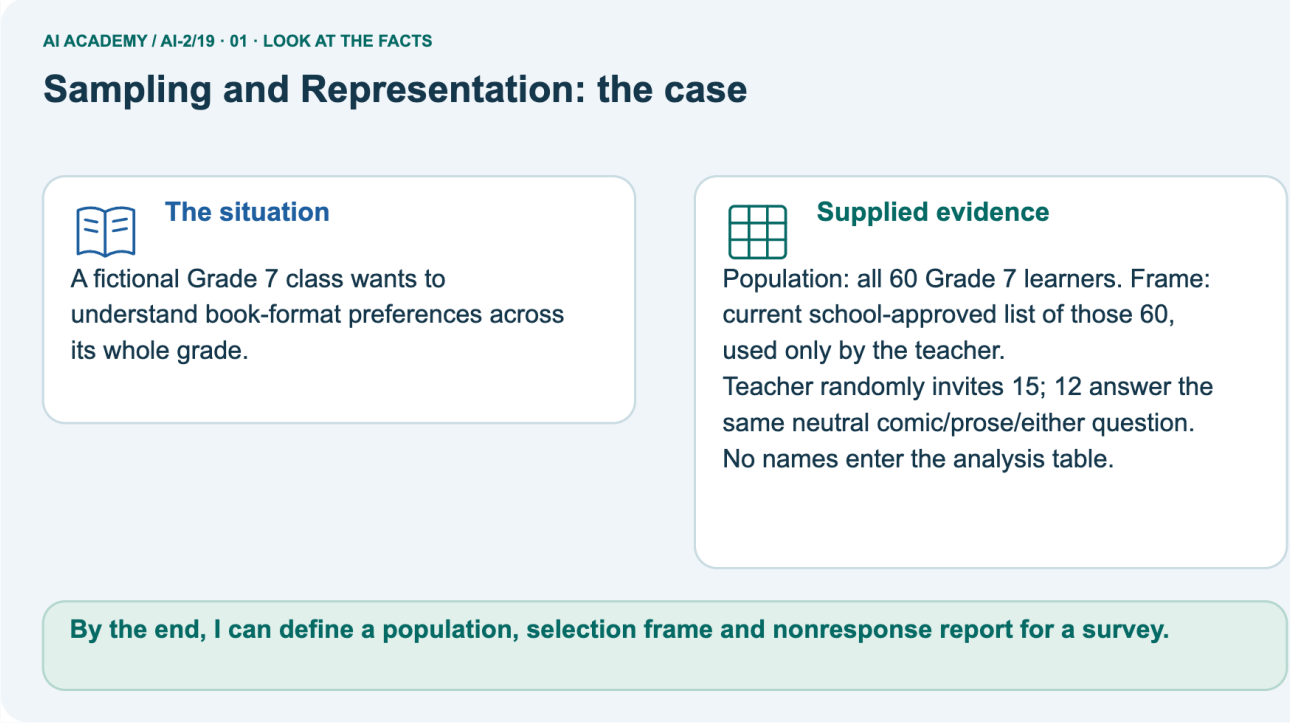
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Sampling and Representation picture from its labels, then record a relevant limit.

01. Sampling and Representation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

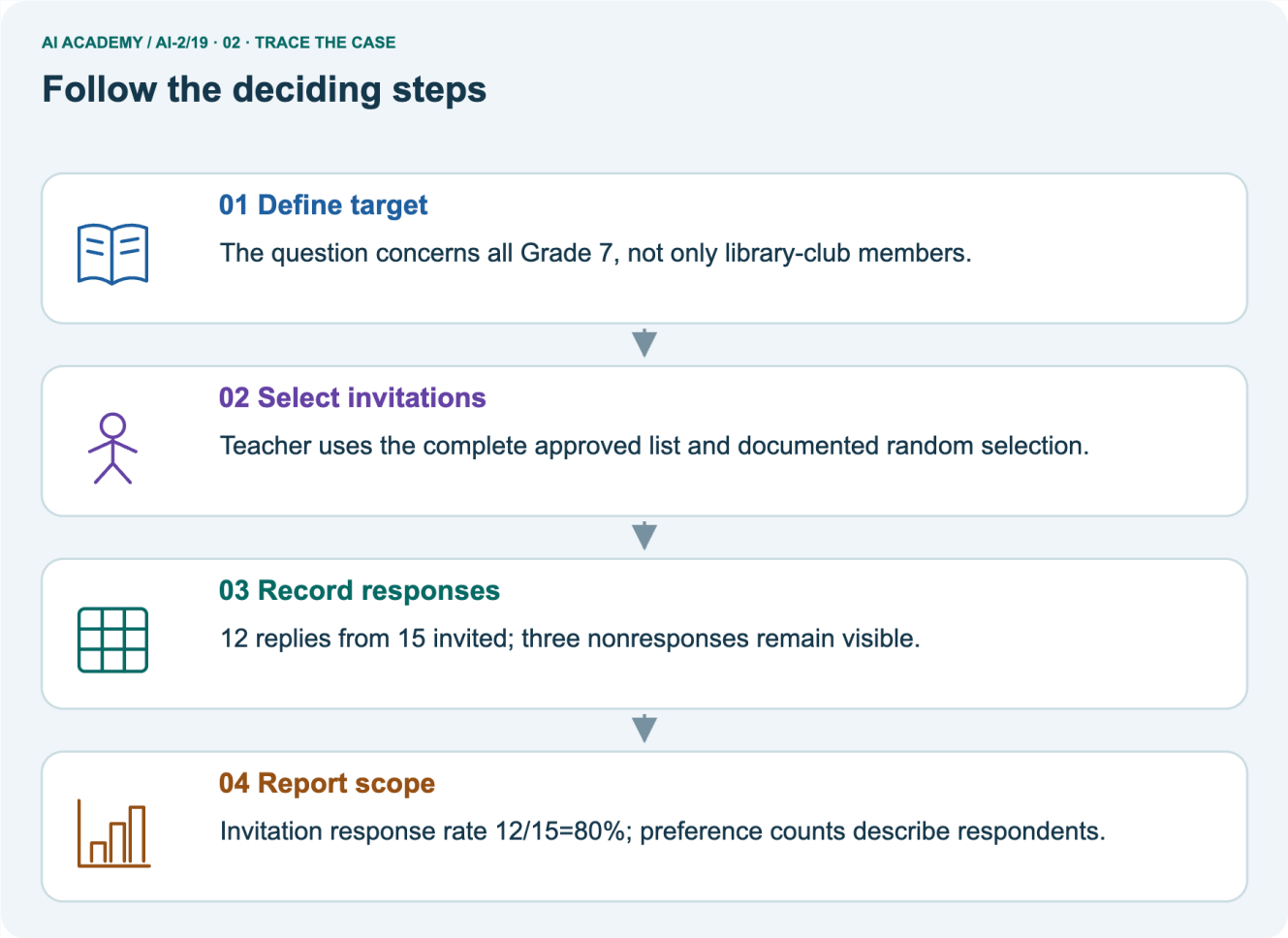
A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Supplied facts: **Population**: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: The question concerns all Grade 7, not only library-club members. → Select invitations: Teacher uses the complete approved list and documented random selection. → Record **response**s: 12 replies from 15 invited; three non**response**s remain visible. → Report scope: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence

AI ACADEMY / AI-2/19 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
Target	All Grade 7 students	Population
Available data	Three club responses	Sample
Observed result	2 of 3 choose Code	Describes respondents

Read each row from left to right and connect the evidence to the stated rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Target | All Grade 7 students | **Population**. Available data | Three club **response**s | **Sample**. Observed result | 2 of 3 choose Code | Describes respondents

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-2/19 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Invitation response rate 12/15=80%; preference counts describe respondents.

Changed condition

Invite only the 15 library-club members instead.

Worked changed-case result

The invitation process no longer covers the whole grade through this frame.
Why: Club members may differ in reading interests; selection is now limited to one group.

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
Changed condition: Invite only the 15 library-club members instead.

Expected result: The invitation **process** no longer covers the whole grade through this frame.
Reason: Club members may differ in reading interests; selection is now limited to one group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-2/19 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

A sample plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined. Use a neutral question and preserve nonresponse coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

What it does not establish

Even the random-invitation plan does not remove nonresponse bias or justify revealing identifiable school lists.

Follow the labelled picture with the matching explanation. Open the vector diagram

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

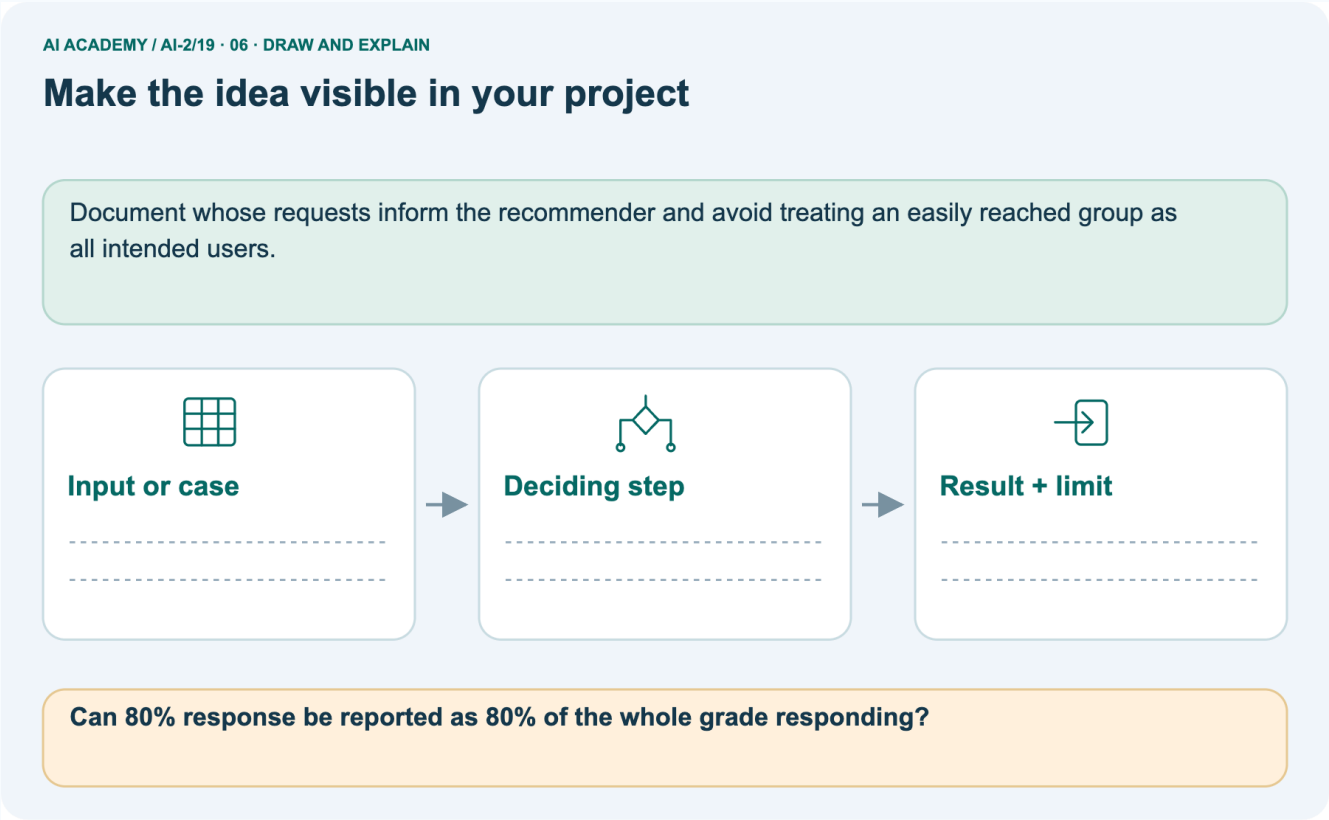
Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Limit: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users. Draw your **input**, deciding step, result and limit.

Supported question: Can **80% response** be reported as **80%** of the whole grade responding?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.